CHAPTER 1

# SOLUTIONS

**CHAPTER ESSENCE AT A GLANCE:**
Quantitative properties of homogeneous liquid solutions, Henry's gas solubility, Raoult's law of vapor pressure, colligative properties, and van 't Hoff factor.

## 4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Henry's Law & Gas Solubility:** The solubility of a gas in a liquid at constant temperature is directly proportional to the partial pressure of the gas above the liquid surface; increasing temperature reduces gas solubility, explaining why aquatic species thrive better in cold water.

• **Raoult's Law & Liquid Mixtures:** For solutions of volatile liquids, the partial vapour pressure of each component equals the product of its pure-component vapour pressure and its mole fraction in solution. Ideal solutions obey this relation across all concentrations with zero enthalpy and volume change on mixing.

• **Deviations & Azeotropic Mixtures:** Positive deviations occur when solute-solvent attractions are weaker than pure component interactions, forming minimum-boiling azeotropes; negative deviations arise from stronger intermolecular attractions like hydrogen bonding, producing maximum-boiling azeotropes.

• **Colligative Properties & Molecular Mass:** Properties including relative lowering of vapour pressure, elevation of boiling point, depression of freezing point, and osmotic pressure depend solely on the number of solute particles rather than their chemical nature.

## ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$p = K_{\mathrm{H}} \cdot x$$

*Henry's Law for gas solubility (K H is Henry's law constant)*

$$p_1 = p_1^{\circ} x_1 \quad \text{and} \quad P_{\text{total}} = p_1^{\circ} + (p_2^{\circ} - p_1^{\circ}) x_2$$

*Raoult's Law for binary volatile liquid mixtures*

$$\frac{p_1^{\circ} - p_1}{p_1^{\circ}} = x_2 = \frac{n_2}{n_1 + n_2} \approx \frac{w_2 M_1}{M_2 w_1}$$

*Relative lowering of vapour pressure (RLVP colligative property)*

$$\Delta T_b = K_b \cdot m = \frac{1000 \cdot K_b \cdot w_2}{M_2 \cdot w_1}$$

*Elevation of boiling point (K b is ebullioscopic constant)*

## CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Osmotic pressure is preferred for polymer molar mass determination because measurements are made at room temperature and pressure values are comfortably large even for dilute solutions.

• **Alert:** [Valuation Rubric] State colligative property formula: $\Delta T_b = (1000 \cdot K_b \cdot w_2)$ divided by $(M_2 \cdot w_1)$ .

## HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

### 1. State Henry's law and mention two of its important industrial/biological applications. [Score: 2 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: Statement]** The solubility of a gas in a liquid at a constant temperature is directly proportional to the partial pressure of the gas present above the surface of liquid: p = K H · x. [1 Mark]

• **[Step 2: Applications]** 1. Soda water bottles are sealed under high pressure to increase $CO_2$ solubility. 2. Scuba divers use air diluted with helium (11.7% He) to avoid decompression sickness ('bends'). [1 Mark]

• **Key Exam Result:** p = K H x; Applications: soda sealing & scuba breathing mix

### 2. State Raoult's law for solutions containing volatile liquids. Write the mathematical form. [Score: 2 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: Statement]** For a solution of volatile liquids, the partial vapour pressure of each component in the solution is directly proportional to its mole fraction: $p_1 = p_1^\circ x_1$ and $p_2 = p_2^\circ x_2$. [1 Mark]

• **[Step 2: Total Pressure]** Total vapour pressure $P = p_1 + p_2 = p_1^\circ x_1 + p_2^\circ x_2 = p_1^\circ + (p_2^\circ - p_1^\circ) x_2$. [1 Mark]

• **Key Exam Result:** P total = $p_1^\circ x_1 + p_2^\circ x_2$

### 3. Differentiate between ideal and non-ideal solutions with one example of each. [Score: 3 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: Ideal Solutions]** Obey Raoult's law over entire range of concentration; $\Delta H$ mix = 0, $\Delta V$ mix = 0; A-B interactions equal A-A and B-B (e.g. n-hexane + n-heptane, benzene + toluene). [1.5 Marks]

• **[Step 2: Non-Ideal Solutions]** Deviate from Raoult's law; $\Delta H$ mix $\neq$ 0, $\Delta V$ mix $\neq$ 0; A-B interactions differ from pure components (e.g. ethanol + acetone, chloroform + acetone). [1.5 Marks]

• **Key Exam Result:** Ideal ($\Delta H$=0, $\Delta V$=0) vs Non-ideal ($\Delta H \neq 0$, $\Delta V \neq 0$)

### 4. Explain positive and negative deviations from Raoult's law with vapour pressure curves. [Score: 3 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: Positive Deviation]** A-B forces < A-A and B-B forces; P total > P calc; $\Delta H$ mix > 0, $\Delta V$ mix > 0 (e.g. ethanol + water; forms minimum boiling azeotrope). [1.5 Marks]

• **[Step 2: Negative Deviation]** A-B forces > A-A and B-B forces (H-bonding); P total < P calc; $\Delta H$ mix < 0, $\Delta V$ mix < 0 (e.g. chloroform + acetone; forms maximum boiling azeotrope). [1.5 Marks]

• **Key Exam Result:** Positive (weaker attraction) vs Negative (stronger H-bonding)

### 5. Derive the relation between relative lowering of vapour pressure and molar mass of solute. [Score: 3 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: Derivation]** By Raoult's law: $(p_1^\circ - p_1)$ divided by $p_1^\circ = x_2 = n_2$ divided by $(n_1 + n_2)$. [1 Mark]

• **[Step 2: Dilute Solution]** For dilute solutions $n_2 << n_1$ implies $x_2$ approx. $n_2$ divided by $n_1 = (w_2$ divided by $M_2)$ divided by $(w_1$ divided by $M_1)$. $(p_1^\circ - p_1)$ divided by $p_1^\circ = (w_2 M_1)$ divided by $(M_2 w_1)$ implies $M_2 = (w_2 M_1 p_1^\circ)$ divided by $[w_1 (p_1^\circ - p_1)]$. [2 Marks]

• **Key Exam Result:** $M_2 = (w_2 M_1 p_1^\circ)$ divided by $[w_1 (p_1^\circ - p_1)]$

### 6. Derive the relation between elevation of boiling point and molality of solution \ΔT b = K b m. [Score: 3 Marks]

**Answer & Key Valuation Steps:**

• **[Step 1: Derivation]** Boiling point elevates when vapour pressure equals 1 atm: $\Delta T$ b proportional to m implies $\Delta T$ b = K b m. [1 Mark]

• **[Step 2: Molality Substitution]** m = $(w_2$ divided by $M_2) \times (1000$ divided by $w_1)$ implies $\Delta T$ b = K b × $(w_2 \times 1000)$ divided by $(M_2 \times w_1)$. $M_2 = (1000 \times K\ b \times w_2)$ divided by $(\Delta T\ b \times w_1)$. [2 Marks]

• **Key Exam Result:** $\Delta T$ b = K b m; $M_2 = (1000$ K b $w_2)$ divided by $(\Delta T$ b $w_1)$

**7. Derive the relation between depression of freezing point and molality of solution \ΔT f = K f m. [Score: 3 Marks]**

**Answer & Key Valuation Steps:**

• **[Step 1: Derivation]** Freezing point depresses due to lowering of vapour pressure: ΔT f proportional to m implies ΔT f = K f m. [1 Mark]

• **[Step 2: Molality Substitution]** m = ($w_2$ divided by $M_2$) × (1000 divided by $w_1$) implies ΔT f = (1000 × K f × $w_2$) divided by ($M_2$ × $w_1$). $M_2$ = (1000 × K f × $w_2$) divided by (ΔT f × $w_1$). [2 Marks]

• **Key Exam Result:** ΔT f = K f m; $M_2$ = (1000 K f $w_2$) divided by (ΔT f $w_1$)